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BANKING MODERNIZATION

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Neeraj Aggarwal
Author

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Building AI-Ready Legacy Systems: An Engineering Framework for Modernization

By Neeraj Aggarwal, Senior Project Manager, Infosys. LinkedIn neeraj.trac3@gmail.com

Introduction

Artificial Intelligence (AI) is rapidly reshaping industries, but integrating it into mission-critical legacy systems remains one of the toughest engineering challenges. While AI has proven its value in customer-facing applications like chatbots and fraud detection, embedding it into the transaction engines and compliance workflows that power financial institutions — or similar legacy platforms in telecom, aerospace, and embedded systems — is far less straightforward.

Legacy infrastructures, often built decades ago on monolithic architectures, resist the flexibility AI requires. Engineers face the dual pressure of maintaining reliability while enabling real-time analytics, automation, and intelligent decisioning. The urgency is clear: rising customer expectations, competition from digital-native players, and regulatory demands all push organizations toward modernization.

This article explores the engineering barriers and enablers of AI adoption in legacy environments, drawing lessons from core banking modernization that apply broadly across industries. By reframing modernization as a systemic engineering problem, we provide a roadmap for practitioners tasked with bridging the gap between yesterday's infrastructure and tomorrow's intelligent systems.

Barriers to AI Integration in Legacy Systems

Legacy Architectures Many mission-critical platforms — from banking transaction engines to telecom switches — were designed decades ago. Their rigid, batch-processing architectures lack the APIs and modularity needed for AI plug-ins. Engineers attempting integration often encounter brittle codebases and limited real-time data access.

Data Fragmentation AI thrives on clean, unified datasets. Yet legacy environments are notorious for fragmented sources, inconsistent metadata, and siloed storage. Without robust

data pipelines, models cannot be trained or deployed reliably. Engineers must spend disproportionate effort on data cleansing before innovation can begin.

Talent Gaps Successful integration requires professionals fluent in both AI and the domain-specific legacy systems. This hybrid skillset is rare. Many failed pilots stem from misaligned solutions designed without deep understanding of the underlying infrastructure.

Reliability and Compliance Concerns In industries like banking, aerospace, or healthcare, reliability is non-negotiable. Engineers hesitate to embed AI into mission-critical workflows

without clear frameworks for explainability, auditability, and fault tolerance. The absence of standardized compliance models slows adoption.

Cultural Resistance Beyond technical hurdles, engineers often face skepticism from operational teams. Concerns about automation displacing jobs or undermining system stability create organizational inertia. Building trust in AI's reliability is as critical as solving the technical challenges.

Engineering Enablers for AI Integration

Leadership Buy-In Successful modernization projects begin with strong engineering leadership. Just as banks benefit from executive sponsorship, engineering teams need clear direction from CTOs, chief architects, or program managers who champion AI adoption. Leadership ensures funding, governance, and alignment across cross-functional teams.

Modular Architecture Rigid monolithic systems are the biggest barrier to AI integration. Engineers who succeed adopt API-first, event-driven designs that allow AI modules to plug into existing workflows without disrupting mission-critical processes. Microservices and containerization provide the flexibility to scale AI components independently, reducing risk and accelerating deployment.

Data Readiness AI cannot thrive without clean, unified data. Engineers must invest in data lakes, lineage tracking, and governance frameworks to ensure models are trained on reliable inputs. Projects that prioritize data quality — through metadata management and standardized schemas — consistently achieve faster AI adoption.

Strategic Partnerships No engineering team operates in isolation. Collaborations with hyperscalers, fintechs, and universities provide access to accelerators, specialized talent, and cutting-edge tools. These partnerships reduce development cycles and bring fresh perspectives to legacy modernization challenges.

AI Governance Trust is essential in mission-critical environments. Engineers must establish model risk management frameworks, audit trails, and ethical AI boards to ensure compliance and reliability. Governance structures not only satisfy regulators but also build confidence among operational teams that AI can be safely embedded into core workflows.

AI Adoption Barriers vs. Success Factors

Category

Barriers

Success Factors

Legacy Systems

Rigid, monolithic architectures; lack of APIs; outdated technologies (COBOL, batch systems)

Modular, API-first, event-driven architectures; microservices and containers

Data

Inconsistent models; poor metadata; fragmented sources

Investment in data lakes, metadata management, lineage tracking, governance

Talent

Shortage of AI-literate professionals with domain expertise

Cross-functional teams; hybrid talent development initiatives

Governance / Compliance

Unclear regulations; lack of explainability and auditability; risk aversion

AI governance frameworks; ethical boards; model risk management

Culture

Fear of automation; job displacement concerns; skepticism toward AI reliability

Internal evangelism; leadership support; transparency and reliability

The AI Adoption Maturity Ladder: An Engineering Roadmap

Integrating AI into legacy systems is not a single event — it's a staged journey. Based on lessons from core banking modernization, we propose the AI Adoption Maturity Ladder, a framework that engineers can apply across mission-critical industries.

Stage 1: Exploration Teams run pilots and proof-of-concepts, often in isolated environments. The focus is on experimentation — testing models, validating feasibility, and building initial confidence.

Stage 2: Integration AI modules are embedded into select workflows. Engineers begin connecting models to production data streams, often through APIs or middleware. Reliability and latency become key concerns.

Stage 3: Modernization Legacy components are replaced with AI-ready modules. This stage requires architectural redesign — moving from monolithic systems to microservices,

containerization, and event-driven frameworks. Engineers must balance modernization with system stability.

Stage 4: Governance Formal oversight structures are established. Engineers implement audit trails, explainability frameworks, and compliance checks. In industries like aerospace or healthcare, this stage is critical for regulatory approval and operational trust.

Stage 5: Optimization AI becomes a continuous improvement engine. Models are retrained, pipelines are refined, and feedback loops are automated. Engineers focus on scalability, resilience, and cross-domain applications.

The ladder provides a structured roadmap for engineers facing the challenge of embedding AI into legacy systems. It highlights that success is not just about algorithms — it's about architecture, governance, and organizational alignment.

Case Study Highlights: Engineering Lessons from Core Banking

Legacy Integration Challenge In one modernization program, a global bank attempted to embed AI into its transaction engine built on COBOL. The rigid batch-processing architecture resisted real-time analytics. Engineers solved this by introducing API gateways and middleware layers, enabling AI modules to access transaction data without rewriting the entire system. **Lesson for Engineers:** When faced with brittle legacy code, build integration bridges rather than attempting wholesale replacement.

Data Readiness Transformation Another institution invested heavily in data lakes and lineage tracking before deploying AI. By standardizing metadata and unifying fragmented sources, engineers created a reliable foundation for model training. **Lesson for Engineers:** Prioritize data infrastructure early — AI adoption fails without clean, consistent inputs.

Talent and Domain Expertise A modernization initiative stalled until engineers with dual expertise in AI and banking systems were brought in. Their ability to align models with domain-specific workflows turned failed pilots into successful deployments. **Lesson for Engineers:** Recruit or train hybrid talent who understand both AI and the legacy domain.

Governance and Trust One North American bank established an AI governance board with audit trails and explainability frameworks. This reassured compliance teams and accelerated adoption. Lesson for Engineers: Build trust frameworks — reliability and transparency are as critical as technical performance.

Cross-Industry Relevance While these examples come from banking, the engineering lessons apply broadly. Telecom networks, aerospace control systems, and healthcare platforms face similar challenges: legacy rigidity, data silos, and compliance demands. The solutions — modular architecture, data readiness, hybrid talent, and governance — are universal.

Conclusion & Recommendations

Artificial Intelligence has the potential to revolutionize mission-critical legacy systems, but adoption is gated by architectural rigidity, fragmented data, and organizational inertia. The lessons from core banking modernization reveal that success is not simply about deploying algorithms — it is about engineering systemic change.

For engineers tackling modernization projects in banking, telecom, aerospace, or healthcare, the roadmap is clear:

- Establish cross-functional AI task forces to align technical and business priorities.
- Invest in data infrastructure and governance to ensure reliable inputs for models.
- Adopt modular, API-driven architectures that allow AI components to integrate without destabilizing legacy systems.
- Implement trust frameworks — audit trails, explainability, and compliance checks — to build confidence among stakeholders.
- Engage external partners to accelerate innovation and fill talent gaps.

AI adoption is not a bolt-on solution; it is a staged journey requiring leadership, architecture, and governance. By treating AI as a strategic enabler of modernization, engineers can transform legacy platforms into intelligent, resilient systems that meet the demands of the digital era.

About Author:

Neeraj Aggarwal is a Senior Project Manager at Infosys with more than two decades of experience leading large-scale modernization programs across global banks and financial institutions. He specializes in AI-driven core banking transformation, cloud migration, and payment gateway modernization, and has been recognized with multiple industry awards for delivering complex, mission-critical change initiatives. Neeraj is a contributor to the Banking Industry Architecture Network (BIAN), a peer reviewer for international IEEE conferences, and a Fellow of the Institution of Electronics and Telecommunication Engineers, reflecting his deep engagement with both enterprise practice and the wider technology community

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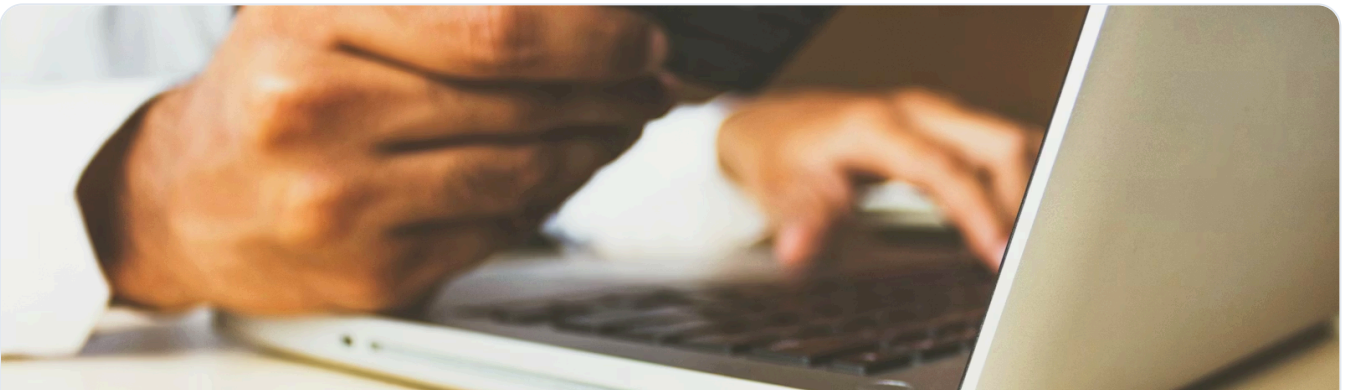
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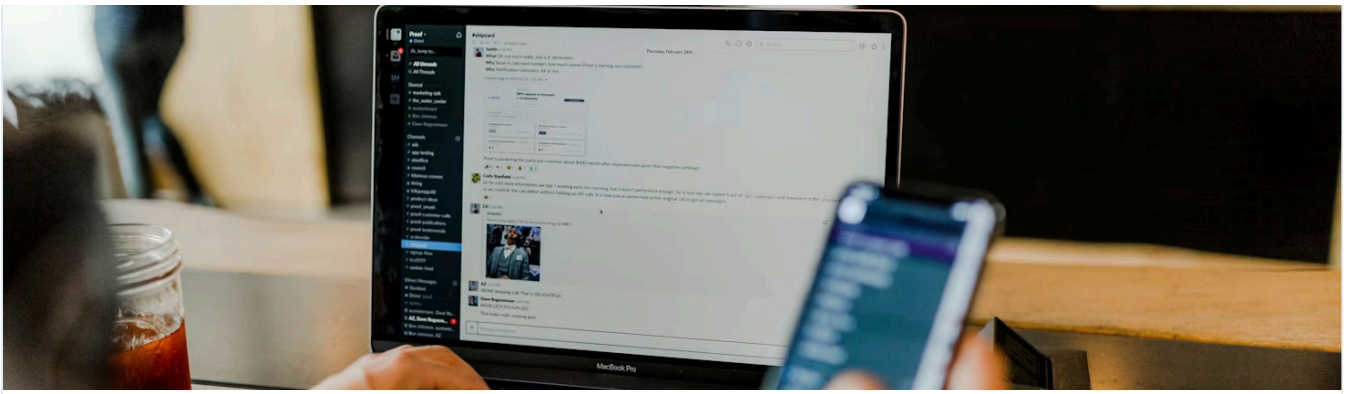
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